# Regular Polyhedra under Typed Observation Filters

## Abstract incidence, realization symmetry, Petrials, and identifiability

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**Status.** This document supersedes the bilingual working note v1.0 and is an application-layer reference migrated to GO Core v0.2. It does not introduce a new polyhedron, classification, realization theorem, or symmetry algorithm. Its contribution is a typed separation of abstract incidence, geometric realization, admissibility, information loss, exact symmetry, and finite-resolution inference.

**Abstract**

The phrase "regular polyhedron" changes meaning when convex polyhedra, star polyhedra, abstract maps, or discrete skeletal realizations are placed in the same discussion. This reference fixes the object class before counting. An abstract polyhedron is a ranked poset satisfying the diamond condition and strong flag connectivity; abstract regularity means transitivity of $\mathrm{Aut}(\mathcal{P})$ on flags. A geometric realization $\beta$ has a generally smaller Euclidean symmetry group $G_\beta$, so an abstractly regular object need not be geometrically regular in a chosen realization. Observation specifications are defined on full subgroupoids of realized objects. Admissibility enlargement and informational coarsening are two different preorders: $\mathcal{C}_1 \subseteq \mathcal{C}_2$ enlarges the admitted domain, whereas $Q_{\text{coarse}} = \kappa \circ Q_{\text{fine}}$ loses distinctions. They cannot be compressed into an undefined inclusion of tuples. A cube and its Petrial give an exact non-injectivity control: the vertex-edge skeleton is unchanged, while six square faces are replaced by four Petrie hexagons and the Euler characteristic changes from $2$ to $0$. Under bounded noise, approximate symmetry residuals and a half-separation criterion give conditional identifiability, not proof of exact symmetry. Counts such as five Platonic solids or $48 = 18+30$ discrete regular skeletal polyhedra in $\mathbb{R}^3$ are therefore convention-scoped results, not values of one universal observation-independent enumeration.

# Contents

# 1 Scope and the category of objects

**Definition 1.1** (Abstract polyhedron)**.** A rank-$3$ abstract polyhedron $\mathcal{P}$ is a partially ordered set with a rank map to $\{-1, 0, 1, 2, 3\}$, unique least and greatest faces, and the following properties.

(i) Every maximal chain, or *flag*, contains exactly one face of each rank.

(ii) If $F < G$ have ranks $i-1$ and $i+1$, exactly two rank-$i$ faces lie strictly between them (diamond condition).

(iii) Every section is flag-connected, equivalently the usual strong flag-connectivity condition holds.

The proper faces of ranks $0, 1, 2$ are vertices, edges, and faces. A reduced flag is the incident triple $(v, e, f)$.

The tuple $(V, E, F, I)$ from the legacy note is only bookkeeping; by itself it does not impose thinness, connectedness, rank, or section axioms. Those axioms are required before standard results about polyhedra and flags are used.

**Definition 1.2** (Realized-polyhedron groupoid)**.** Let $\mathbf{RPol}_3$ be the groupoid whose objects are

$$X = (\mathcal{P}, \beta, \mathsf{T}),$$

where $\mathcal{P}$ is an abstract polyhedron, $\beta$ is a declared realization in $\mathbb{R}^3$, and $\mathsf{T}$ records the realization convention: straight or other edges, planar, skew, star, zig-zag, or helical faces, discreteness, local finiteness, and faithfulness. An arrow $X \to X'$ is a rank-preserving incidence isomorphism $\varphi : \mathcal{P} \to \mathcal{P}'$ together with an ambient isometry $a$ intertwining the realizations and all declared edge and face data.

The convention $\mathsf{T}$ is part of the type. A convex face-lattice realization, a regular map, and a skeletal apeirohedron are not silently treated as objects of one unqualified category.

# 2 Flags, abstract regularity, and string C-groups

By the diamond condition, every flag $\Phi$ has a unique $i$-adjacent flag $\Phi^i$, differing only in its rank-$i$ face, for $i = 0, 1, 2$. The resulting edge-coloured flag graph is connected.

**Definition 2.1** (Abstract regularity)**.** The automorphism group $\Gamma(\mathcal{P}) = \mathrm{Aut}(\mathcal{P})$ consists of all rank-preserving order automorphisms. The abstract polyhedron is regular when $\Gamma(\mathcal{P})$ acts transitively on $\mathrm{Flag}(\mathcal{P})$.

**Proposition 2.2** (Simple flag transitivity)**.** *For an abstract polyhedron, an automorphism fixing one flag is the identity. Hence a regular abstract polyhedron has a simply transitive flag action. If it is finite, then*

$$|\Gamma(\mathcal{P})| = |\mathrm{Flag}(\mathcal{P})| = 4f_1, \tag{1}$$

*where* $f_1 = |E|$.

*Proof.* Rank preservation fixes every member of the chosen flag. The diamond condition then fixes each adjacent flag, and strong flag connectivity propagates this to all flags and faces. Thus a flag stabilizer is trivial. Each edge has two incident vertices and two incident faces, so it belongs to four reduced flags. □

Fix a base flag. For a regular $\mathcal{P}$, let $\rho_i$ be the unique automorphism sending it to its $i$-adjacent flag. For type $\{p, q\}$,

$$\rho_i^2 = 1, \qquad (\rho_0\rho_2)^2 = 1, \qquad \text{ord}(\rho_0\rho_1) = p, \qquad \text{ord}(\rho_1\rho_2) = q, \tag{2}$$

and the intersection property is

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_i : i \in J \rangle = \langle \rho_i : i \in I \cap J \rangle. \tag{3}$$

These data form a rank-$3$ string C-group. The orders $p, q$ are exact, and additional quotient relations may be present. Therefore $\{p, q\}$ is local type data, not a complete global identifier.

# 3 Abstract symmetry versus realization symmetry

For a faithful nondegenerate realization $\beta$, define

$$G_\beta = \{a \in \text{Isom}(\mathbb{R}^3) : a \text{ preserves the realized vertices, edges, and declared faces}\}.$$

Its action induces a subgroup $\iota_\beta(G_\beta) \leq \text{Aut}(\mathcal{P})$.

**Definition 3.1** (Geometric regularity)**.** The realization $(\mathcal{P}, \beta, \mathsf{T})$ is geometrically regular when $G_\beta$ is transitive on its geometric flags.

**Proposition 3.2** (One-way implication)**.** *A faithful geometrically regular realization has an abstractly regular incidence polyhedron. The converse fails.*

*Proof.* The induced subgroup of $\text{Aut}(\mathcal{P})$ is already flag-transitive, so the full automorphism group is flag-transitive. Conversely, give the abstract cube three pairwise distinct edge lengths in orthogonal directions. Its incidence automorphism group still has order $48$ and is flag-transitive, but the rectangular cuboid has only the eight independent coordinate sign changes. Edge directions cannot be permuted, leaving six geometric flag orbits. □

Thus the legacy symbol $G(\mathcal{P})$, described as "geometric or combinatorial depending on the setting", is ill-typed. The two groups and the homomorphism between them must be named.

# 4 Observation specifications and two preorders

**Definition 4.1** (Typed observation specification)**.** An observation specification is

$$\mathcal{O} = (\mathcal{C}_\mathcal{O}, Y_\mathcal{O}, Q_\mathcal{O}),$$

where $\mathcal{C}_\mathcal{O}$ is a full subgroupoid of $\mathbf{RPol}_3$, stable under isomorphism, $Y_\mathcal{O}$ is a declared data or label space, and

$$Q_\mathcal{O} : \text{Ob}(\mathcal{C}_\mathcal{O}) \longrightarrow Y_\mathcal{O}$$

is constant on the isomorphisms the protocol treats as nuisance transformations.

There are two independent comparisons.

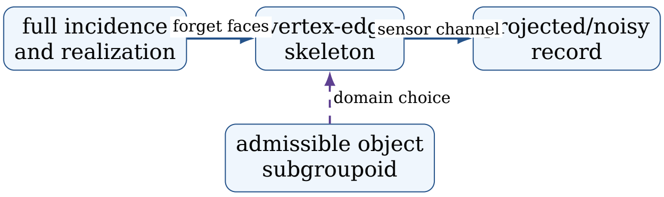
$$\mathcal{O}_1 \preceq_{\mathrm{adm}} \mathcal{O}_2 \iff \mathcal{C}_{\mathcal{O}_1} \text{ is a full subgroupoid of } \mathcal{C}_{\mathcal{O}_2}, \tag{4}$$
$$\mathcal{O}_{\mathrm{fine}} \succeq_{\mathrm{info}} \mathcal{O}_{\mathrm{coarse}} \iff Q_{\mathrm{coarse}} = \kappa \circ Q_{\mathrm{fine}} \text{ on a common domain} \tag{5}$$

for some deterministic $\kappa$. Then $\operatorname{Ker} Q_{\mathrm{fine}} \subseteq \operatorname{Ker} Q_{\mathrm{coarse}}$: the coarse channel identifies at least as many objects.

**Theorem 4.2** (Scoped filter monotonicity)**.** *Let $R$ be one fixed predicate, such as abstract regularity or geometric regularity under one declared convention. If $\mathcal{O}_1 \preceq_{\mathrm{adm}} \mathcal{O}_2$, then*
$$\{X \in \mathrm{Ob}(\mathcal{C}_{\mathcal{O}_1}) : R(X)\} \subseteq \{X \in \mathrm{Ob}(\mathcal{C}_{\mathcal{O}_2}) : R(X)\}.$$
*No ordering of information kernels follows without equation* (5)*, and no count comparison follows if different quotient equivalences are used.*

*Proof.* The set inclusion is immediate from inclusion of the object domains while the predicate is held fixed. The remaining claims concern different data and therefore require a factorization map $\kappa$; domain inclusion alone supplies none. □

*Boundary* 4.3 (The historical family is not a chain)*.* Removing finiteness while retaining planarity and allowing self-intersection while retaining finiteness relax different predicates. The resulting admissible classes may be incomparable. The old "filter ladder" is therefore replaced by a filter poset.




The horizontal arrows are lossy channels. The vertical arrow is a choice of domain. Calling both operations a "filter" without types caused the legacy order error.

# 5 Classical spherical count and its hypotheses

Let a finite equivelar polyhedral map of type $\{p, q\}$ have $f$-vector $(V, E, F)$. Double counting incidences gives
$$pF = 2E, \qquad qV = 2E. \tag{6}$$
If the underlying surface is the sphere,
$$V - E + F = 2, \tag{7}$$
and therefore
$$\frac{1}{p} + \frac{1}{q} - \frac{1}{2} = \frac{1}{E} > 0. \tag{8}$$
For integers $p, q \geq 3$, this leaves precisely
$$(p, q) \in \{(3,3), (4,3), (3,4), (5,3), (3,5)\}.$$
This proves the list of possible local types under the stated spherical, finite, thin, equivelar hypotheses. Existence and uniqueness up to similarity of the corresponding convex regular realizations are the separate classical realization theorem. Equation (8) is not a

Table 1: Exact finite controls for the five convex regular types.

| Object | $p$ | $q$ | $V$ | $E$ | $F$ | $\lvert\text{Flag}\rvert = 4E$ |
|---|---|---|---|---|---|---|
| Tetrahedron | 3 | 3 | 4 | 6 | 4 | 24 |
| Cube | 4 | 3 | 8 | 12 | 6 | 48 |
| Octahedron | 3 | 4 | 6 | 12 | 8 | 48 |
| Dodecahedron | 5 | 3 | 20 | 30 | 12 | 120 |
| Icosahedron | 3 | 5 | 12 | 30 | 20 | 120 |

proof classifying star polyhedra, higher-genus maps, infinite tilings, or skeletal apeirohedra.

Duality reverses the face order:

$$\mathcal{P} \longmapsto \mathcal{P}^*, \qquad (V, E, F) \longmapsto (F, E, V), \qquad \{p, q\} \longmapsto \{q, p\},$$

and preserves the automorphism-group order. Duality is an exact combinatorial operation, not an optical viewpoint change.

# 6 Skeleton non-identifiability and the Petrial control

For a regular map with distinguished generators $(\rho_0, \rho_1, \rho_2)$, the Petrie operation replaces the generating triple by

$$(\rho_0\rho_2, \rho_1, \rho_2). \tag{9}$$

Because $\rho_0$ and $\rho_2$ commute and are involutions, $\rho_0\rho_2$ is an involution. On the map, the operation keeps vertices and edges and selects Petrie polygons as faces.

**Proposition 6.1** (Explicit skeleton non-injectivity)**.** *Let $Q_{\text{skel}}(\mathcal{P}) = (V, E)$. The cube $\mathcal{P}_\square$ and its Petrial $\mathcal{P}_\square^\pi$ satisfy*

$$Q_{\text{skel}}(\mathcal{P}_\square) = Q_{\text{skel}}(\mathcal{P}_\square^\pi), \qquad \mathcal{P}_\square \ncong \mathcal{P}_\square^\pi.$$

*The first has six square faces and Euler characteristic $2$; the second has four Petrie hexagons and Euler characteristic $0$.*

*Proof.* Both structures use the eight cube vertices and twelve cube edges. The six square cycles give $4 \cdot 6 = 24 = 2E$ edge-face incidences. The four length-six Petrie cycles also give $6 \cdot 4 = 24 = 2E$. Their $f$-vectors are respectively $(8, 12, 6)$ and $(8, 12, 4)$, so they cannot be isomorphic as ranked posets and have Euler characteristics $2$ and $0$. □

Thus "same skeleton, different face observation" is now an exact counterexample, not a metaphor. A two-dimensional rendering is even coarser: projection can merge vertices, create apparent crossings, or hide depth order. Those effects belong to the sensor channel, not to the incidence definition.

# 7 Finite-resolution symmetry and identifiability

Let observed vertices be

$$y_i = x_i + \eta_i, \qquad \|\eta_i\| \le \epsilon,$$

after a declared registration, labeling, and incidence-matching protocol. For a candidate isometry $g$ and allowed incidence permutations $\Pi$, define the RMS residual

$$r_Y(g) = \min_{\pi \in \Pi} \left( \frac{1}{n} \sum_{i=1}^{n} \left\| g y_i - y_{\pi(i)} \right\|^2 \right)^{1/2}, \qquad \delta_Y(g) = \frac{r_Y(g)}{\ell_{\text{ref}}}, \tag{10}$$

where $\ell_{\text{ref}} > 0$ is declared.

**Proposition 7.1** (Exact symmetry under bounded noise)**.** *If $g x_i = x_{\pi(i)}$ for one allowed $\pi$, then*

$$r_Y(g) \le 2\epsilon.$$

*The converse is false: a small residual certifies only threshold-dependent approximate symmetry.*

*Proof.* Isometries preserve noise norms, so each residual vector is $g\eta_i - \eta_{\pi(i)}$ and has norm at most $2\epsilon$. The RMS bound follows. Arbitrarily small asymmetric perturbations give nonzero but arbitrarily small residuals. □

Let $d_Q$ be a declared metric on observations modulo nuisance isometries and relabelings, and let $\{M_1, \ldots, M_K\}$ be a finite candidate library. Define

$$\Delta_i = \min_{j \neq i} d_Q(M_i, M_j).$$

**Theorem 7.2** (Half-separation certificate)**.** *If $\Delta_i > 0$ and $d_Q(Y, M_i) < \Delta_i/2$, then $M_i$ is the unique nearest candidate.*

*Proof.* For $j \neq i$, the triangle inequality gives

$$d_Q(Y, M_j) \ge d_Q(M_i, M_j) - d_Q(Y, M_i) > \Delta_i/2 > d_Q(Y, M_i).$$

□

This is conditional identifiability inside a declared finite library. It does not show that the true object belongs to that library.

For a finite flag set and a declared group $G$, let its flag orbits have weights $w_j$. The diagnostic

$$H^{(2)}_{\text{orb}}(G) = -\sum_j w_j \log_2 w_j \tag{11}$$

vanishes exactly when there is one orbit. It is a combinatorial Shannon diagnostic, not thermodynamic entropy, and it is undefined for an infinite flag set without a probability measure.

# 8 Convention-scoped taxonomy and claim firewall

*Boundary* 8.1 (Claim firewall). This reference does not claim a new classification, a new regular polyhedron, completeness beyond a cited convention, or recovery of incidence from a picture. Its strict internal results are the typed filter preorders, the separation of abstract and geometric symmetry, the explicit cube–Petrial non-injectivity control, and the finite-resolution identification bounds.

# 9 Protocol minimum and benchmark audit

A reproducible polyhedral observation claim must declare:

Table 2: Counts are meaningful only after the object convention is fixed.

| Count | Object convention | Permitted interpretation |
|---|---|---|
| 5 | finite convex regular polyhedra in $\mathbb{R}^3$, up to similarity | Classical Platonic classification. |
| 4 | classical Kepler–Poinsot regular star polyhedra | Star-face and self-intersection conventions are part of the type. |
| $48 = 18 + 30$ | Schulte's discrete geometrically regular skeletal polyhedra in $\mathbb{R}^3$: 18 finite and 30 infinite | Published classification in that specific skeletal convention. |
| Not 48 | all abstract regular rank-$3$ polytopes or all combinatorially regular realizations | The abstract and realization universes are much larger. |

1. the abstract object axioms and rank;
2. the realization convention T, ambient space, and faithfulness;
3. whether regularity uses Aut$(\mathcal{P})$ or $G_\beta$;
4. the admissible subgroupoid and the equivalence used for counting;
5. the observation map, projection, labeling, and face-selection rule;
6. spatial resolution, noise bound or law, registration group, estimator, threshold, and uncertainty;
7. for infinite structures, discreteness, local finiteness, observation window, and boundary treatment.

The executable companion checks all five Platonic incidence ledgers, the spherical type inequality, dual pairs, all cube square and Petrie cycles, the $48$-element cube action, the $8$-element generic cuboid action, flag-orbit counts, bounded-noise residuals, and half-separation certificates. Passing these controls verifies the claims encoded here; it is not an independent classification proof.

# References


[1] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Encyclopedia of Mathematics and its Applications 92, Cambridge University Press, 2002. doi:10.1017/CBO9780511546686.

[2] P. McMullen and E. Schulte, “Regular Polytopes in Ordinary Space”, *Discrete & Computational Geometry* 17 (1997), 449–478. doi:10.1007/PL00009304.

[3] E. Schulte, “Regular Incidence Complexes, Polytopes, and C-Groups”, 2017. arXiv:1711.02297.

[4] E. Schulte, “Polyhedra, Complexes, Nets and Symmetry”, *Acta Crystallographica A* 70 (2014), 203–216. doi:10.1107/S2053273314000217.

[5] B. Grünbaum, “Regular Polyhedra—Old and New”, *Aequationes Mathematicae* 16 (1977), 1–20. doi:10.1007/BF01836414.

[6] A. W. M. Dress, “A Combinatorial Theory of Grünbaum's New Regular Polyhedra, Part I”, *Aequationes Mathematicae* 23 (1981), 252–265. doi:10.1007/BF02188039.

[7] A. W. M. Dress, “A Combinatorial Theory of Grünbaum's New Regular Polyhedra, Part II: Complete Enumeration”, *Aequationes Mathematicae* 29 (1985), 222–243. doi:10.1007/BF02189831.